

# **Online Medical Pharmacy Management System**

**Course Name: Object-Oriented Programming I Lab**

**Instructor's Name: Tanjina Helaly**

**Affiliation: Assistant Professor, Department of  
Computer Science and Engineering, University of Asia  
Pacific**

**Date of Submission: 18-11-2025**

**Submitted By,**

**14 / 24101014: Saad Al Mujahid**

**17 / 24101017: MD. Mahid Hasan Dewan**

**18 / 24101018: Tajbiha Maisha**

**25 / 24101025: Taohedul Islam**

**26 / 24101026: Rifa Tasfiah Sadia**

## **Table of Content**

|                                                                   |           |
|-------------------------------------------------------------------|-----------|
| <b>1. Objective</b>                                               | <b>3</b>  |
| <b>2. Problem Statement</b>                                       | <b>3</b>  |
| <b>3. Software &amp; Tools Used</b>                               | <b>4</b>  |
| <b>4. OOP Theory / Concepts Applied</b>                           | <b>4</b>  |
| <b>5. Code Implementation</b>                                     | <b>5</b>  |
| <b>Initial Design:</b>                                            | <b>5</b>  |
| <b>6.1 GUI Design</b>                                             | <b>7</b>  |
| <b>6.2 GUI Screenshot from Implemented Project</b>                | <b>8</b>  |
| <b>7. Member Contribution</b>                                     | <b>10</b> |
| <b>8. Conclusion</b>                                              | <b>11</b> |
| <b>8.1 Challenges Faced during Project</b>                        | <b>11</b> |
| <b>8.2 Lesson Learned (Both Technical and Management aspects)</b> | <b>11</b> |

# 1. Objective

The objective of this lab project is to design and implement a comprehensive Java application with the following key components:

## Object-Oriented Programming Concepts

The application will demonstrate fundamental object-oriented programming principles, including:

- **Classes and Objects:** Structuring the program using reusable blueprints and instances.
- **Inheritance:** Establishing hierarchical relationships to promote code reuse and extensibility.
- **Encapsulation:** Protecting data integrity by restricting direct access to class members.
- **Polymorphism:** Enabling methods to behave differently based on the object's type.
- **Abstraction:** Simplifying complex systems by focusing on essential features while hiding details.

## Additional Functional Requirements

In addition to OOP concepts, the project will incorporate:

- **Input/Output Handling:** Managing user inputs and displaying outputs efficiently to ensure smooth program interaction.
- **Exception Handling:** Implementing mechanisms to gracefully detect and manage runtime errors, enhancing program robustness.
- **Graphical User Interface (GUI):** Designing an interactive and user-friendly visual interface to facilitate seamless user interaction with the application.

# 2. Problem Statement

In this project, we need to build a medicine inventory and ordering system using Java.

We must create a program that will help me handle product information such as stock, pricing, expiry dates, and billing.

we must create a console-based program where employees can add and manage medicines, check expiry, search items, and handle order requests, while customers can place orders, view their order details, and pay bills. We also need to handle errors properly using the exceptions, created and allow switching between employee and customer roles.

### **3. Software & Tools Used**

- Programming Language: Java (JDK version)
- IDE: [Eclipse / IntelliJ ]
- OS: [Windows / MacOS]
- Other dependencies (if any)

### **4. OOP Theory / Concepts Applied**

- ☐ Class/Object
- ☐ Inheritance
- ☐ Polymorphism
- ☐ Package
- ☐ Array List
- Abstraction
- Input/Output Handling
- Exception Handling
- Graphical User Interface (GUI)

## 5. Code Implementation

The backend code files has been attached separately.

## Initial Design:

## MedPharma Login

**Role:** ☐Employee ☐Customer

Login

Exit

# Employee Dashboard

Welcome, Employee!

 Add Item

 View & Accept Orders

 Find Expired Medicine

 Log Out

Type

Manufacturer

Name

Search Items

Search results Will Appear here

No active Searches

Clicking a button will open a new sub-form/window

Add Item

Medicine Details

Medicine name :

Manufacturer :

Dose :

Unit :

Unit Price :

Quantity :

Expiration Date : (dd/MM/yyyy).....

Submit

Accessory Details

Accessory Name :

Manufacturer :

Model No :

Unit Price :

Quantity :

Submit

Back

[illegible]

## Expired Medicine List

## Pending Orders

[illegible]

[Back](#)

## Customer Dashboard

Welcome, Valued Customer!

 Search / Order Item

 Pay or View My Orders

Logout

Clicking a button will open a new sub-form/window

[View Or Request Order](#)

Type 

Manufacturer 

Name ▼

Search items

Search Results Will Appear Here....

No active information to display.

Item ID....

Quantity.....

Contact Information....

[Request Order](#)

Enter Order ID To Search Order /PayBill

Order ID.....

[View My Order](#)

My Orders

| Order ID | Item ID | Contact Information | Quantity | Status | Order Date | Delivery Date |
|----------|---------|---------------------|----------|--------|------------|---------------|
|----------|---------|---------------------|----------|--------|------------|---------------|

## Details

**Item: Paracetamol 500mg Tablets**

**Category:** Medicine

**Description:** Pair relief and fever reducer.  
Batch P123. Expiry- 2025-10-31.

**Quantity:** 1500 units.

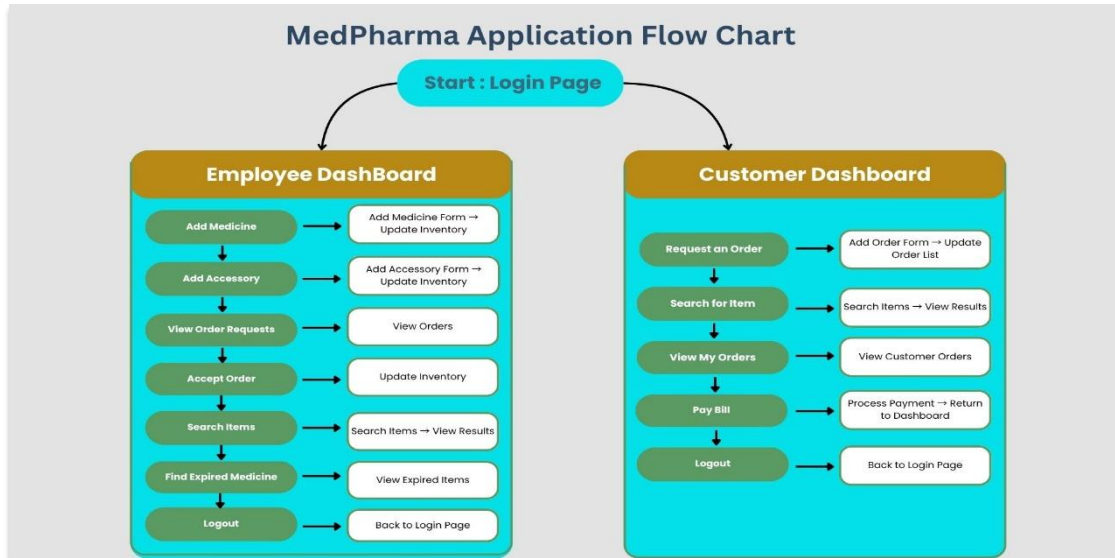
**Supplier:** PharmaCorp

**Notes:** High-demand item.

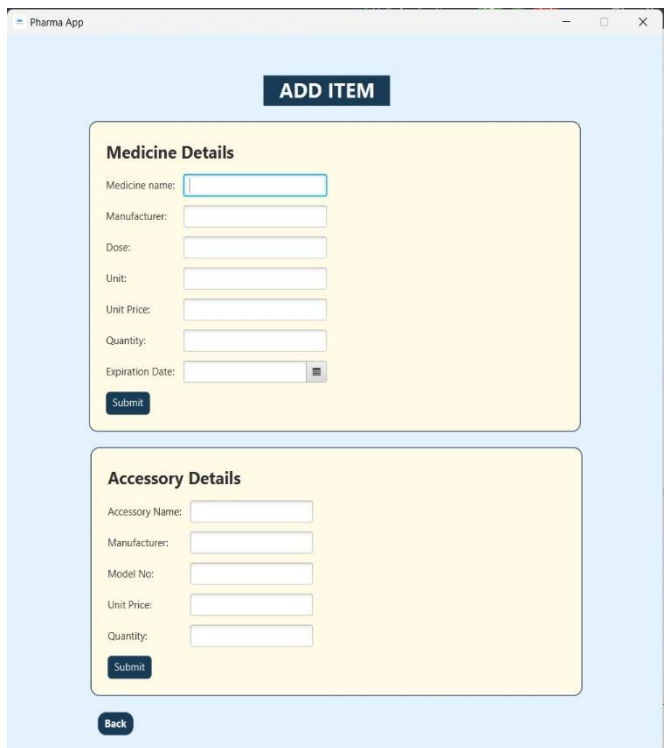
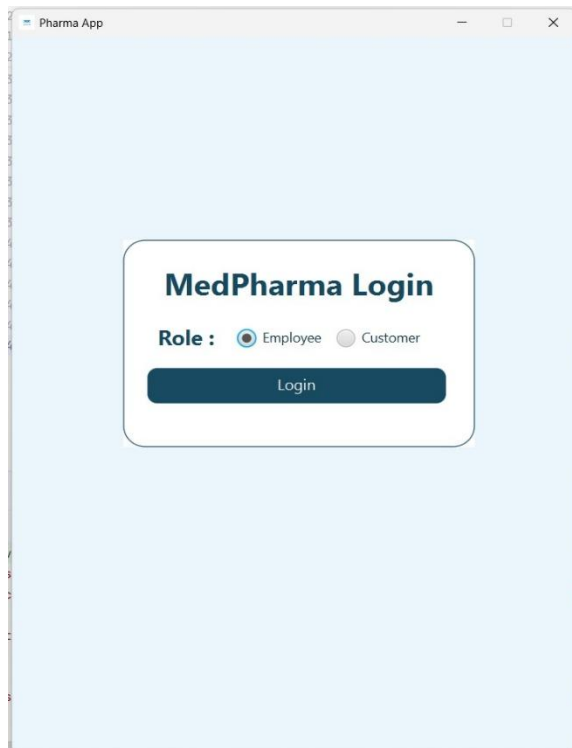
## 6. GUI

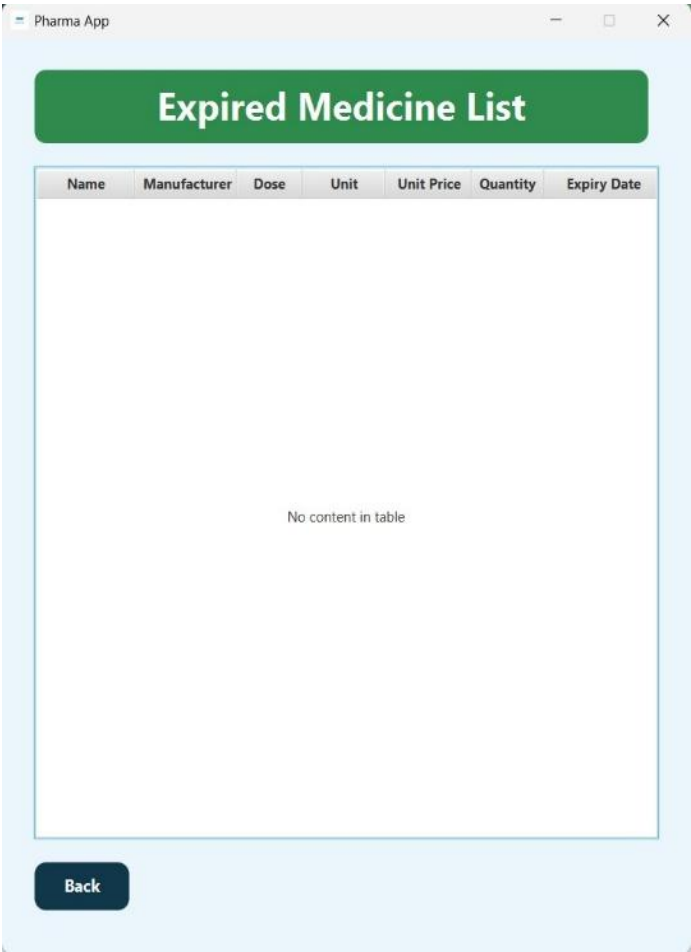
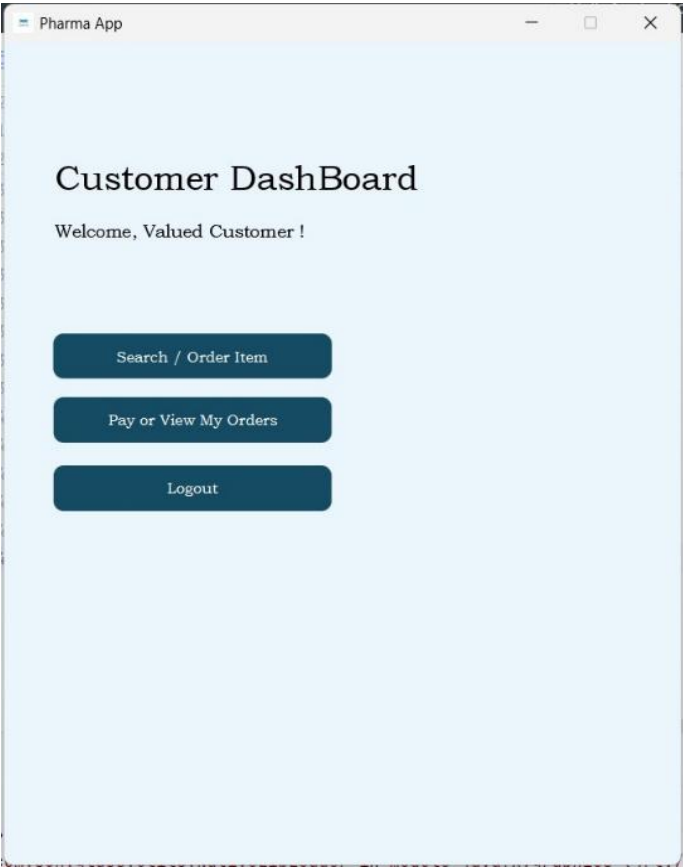
### 6.1 GUI Design

The file of initial and flow diagram is added.



### 6.2 GUI Screenshot from Implemented Project





Pharma App

View Or Request Order

Type

Manufacturer

Name

Item ID....

Quantity...

Contact....

Request Order

Back

Pharma App

Enter Order ID To Accept

Accept Order

Pending Orders

| Order ID            | Item Name | Order Date | Order Status | QTY | Unit Price | Total |
|---------------------|-----------|------------|--------------|-----|------------|-------|
| No content in table |           |            |              |     |            |       |

Back

Pharma App

Enter Contact Information To Search Order /PayBill

View My Order

PayBill

My Orders

| Order ID            | Item ID | Information | Quantity | Status | Order Date | Delivery Date |
|---------------------|---------|-------------|----------|--------|------------|---------------|
| No content in table |         |             |          |        |            |               |

Back

**Update project: No one Can order Expired Medicine Anymore and View/Pay Bill got updated as they now can add directly by selecting the order and then pay bill if order is completed and is not paid yet.**

Combo box and Suggestions were added according to medicine list.

A table is provided to show each member's assigned tasks alongside their completed work. Every participant is responsible for:

| Member ID: Name              | Planned Tasks                               | Completed Tasks                             |
|------------------------------|---------------------------------------------|---------------------------------------------|
| 24101014: Saad Al Mujahid    | Log in page                                 | Helped on Log in Button Design.             |
| 24101017: Mahid Hasan Dewan  | Employee Dashboard, Expired Medicine List   | Employee Dashboard, Expired Medicine List   |
| 24101018: Tajbiha Maisha     | view or Request Order, view or Accept Order | view or Request Order, view or Accept Order |
| 24101025: Taohedul Islam     | Add Item and Accessory, Customer Dashboard. | Add Item and Accessory, Customer Dashboard. |
| 24101026: Rifa Tasfiah Sadia | Log in , Playbill or View Orders            | Log in, Playbill or View Orders             |

## 8. Conclusion

### 8.1 Challenges Faced during Project

- Scene Navigation: The Login as Employee button didn't work at first due to incorrect FXML-controller mapping. Fixing the references solved the issue.
- Back Button: Navigation backwards failed initially because of missing scene-switching logic. Using consistently resolved it.
- UI Design: Making the interface simple yet appealing was tricky. I had to ensure consistent spacing, button styles, and color choices across all pages.
- Backend Integration: Connecting frontend pages with backend features like adding medicines or processing orders required careful data handling and debugging.

### 8.2 Lesson Learned:

#### Technical Lessons

- FXML-Controller Integration

Ensuring proper linkage between FXML files and controllers is vital for smooth UI-logic interaction.

- Scene Transition Management

Using a streamlined navigation, reduced repetitive code, and kept the app modular.

- User-Centered Design

A clean, simple layout with thoughtful colors and buttons improved usability and engagement.

- Robust Error Handling

Managing invalid inputs or missing data created a smoother, more reliable user experience.

#### Management Lessons

- Structured Time Planning

Breaking tasks into smaller steps helped maintain focus and avoid overwhelm.

- Cross-Team Communication

Clear coordination between frontend and backend ensured alignment and prevented conflicts.

- Feature Prioritization

Focusing on essential features first guaranteed core functionality before adding extras.

- Practical Problem-Solving

Debugging UI flaws, missing methods, and integration issues strengthened troubleshooting skills.